

## DAY 18 — PROMPT ENGINEERING (SYSTEM + TOOL CONTROL)

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### GOAL

Upgrade from: Loose prompts ❌

to: Controlled + consistent AI ✅

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### WHAT YOU WILL IMPROVE

1. Final answer prompt
  2. Planner prompt
  3. Tool prompts (Excel / RAG)
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#### ✅ STEP 1 — GLOBAL SYSTEM PROMPT config.py

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##### ✅ Add

```
SYSTEM_PROMPT = """ You are an intelligent AI assistant.
```

Rules:

- Always be precise and clear
  - Prefer structured answers
  - Use step results if provided
  - Do NOT hallucinate
  - If unsure, say "Not enough information"
  - Keep answers concise but complete """
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#### ✅ STEP 2 — FIX FINAL ANSWER PROMPT : app/agent.py

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##### ✅ Replace final prompt

```
from config import SYSTEM_PROMPT
```

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### ✅ Update this part:

```
final_prompt = f"""
```

```
{SYSTEM_PROMPT}
```

```
Context:
```

```
{results}
```

```
User Question:
```

```
{query}
```

```
Instructions:
```

- Use ONLY the context
- Be accurate and structured
- Avoid unnecessary text

```
Answer: """
```

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### ✅ STEP 3 — IMPROVE PLANNER

 app/planner.py

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### ✅ Replace prompt

```
prompt = f"""
```

```
You are a planning agent.
```

```
Break the query into steps.
```

```
Tools:
```

- EXCEL → structured data analysis
- PDF\_TABLE → table data from PDF
- RAG → document-based reasoning

```
Rules:
```

- Use minimal steps
- Use correct tool
- Do NOT repeat steps

Return format:

Step 1: TOOL: task

Query:

{query}"""

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#### ✅ STEP 4 — IMPROVE RAG PROMPT



where you call LLM for context answer

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**Replace with:**

```
rag_prompt = f"""
```

Answer based ONLY on context.

Context:

{context}

Question:

{query}

Rules:

- Do NOT add outside knowledge
- If answer not found, say "Not in context"

Answer: """

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#### ✅ STEP 5 — IMPROVE EXCEL PROMPT



inside ExcelAgent.smart\_query()

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**Replace prompt with:**

```
prompt = f"""
```

You are a pandas expert.

DataFrame: df

Schema:

{schema\_desc}

Rules:

- Return ONLY python code
- No explanations
- Use safe operations

Question:

{question}"""

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### ✅ STEP 6 — OPTIONAL (QUALITY BOOST)

👉 Add **answer formatting**

- Use bullet points if needed
  - Keep structure clean
- 

### ✅ TEST

Run: uvicorn app.api:app --reload

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### ✅ Try:

```
>> what is ai  
>> summarize document  
>> compare trends  
>> average sales
```

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### ✅ EXPECTED IMPROVEMENT

Before:

Random answers

Inconsistent style

After:

Structured 

Accurate 

Consistent 

Less hallucination 

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### FINAL FLOW NOW

User Query

↓

Planner (optimized)

↓

Parallel tools

↓

Results

↓

Controlled final prompt 

↓

Consistent Answer

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### WHAT YOU BUILT TODAY

Prompt control layer

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### SYSTEM LEVEL NOW

You now have:

 AI system

 API + Docker

 Async execution

 Caching

✓ ✓ Prompt optimization (NEW)

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✓ **INDUSTRY TERM**

Prompt Engineering Layer

Used in:

- ChatGPT
  - Copilot
  - Production LLM systems
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✓ **NEXT**

DAY 19 → Advanced Memory (long-term + vector memory)

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If ready: 👉 say **Day 19**